Portfolio.

Omar Saif
Mechanical Design Engineer

Think.
Make.
Break.
Repeat.

Design Philosophy

Pioneered by Otto von Guericke at the start of 20th century, the iterative design process has been the cornerstone of many modern design language for various industries across the spectrum.

Think, Make, Break, Repeat – 4 steps of iterative design thinking that emphasizes on rapid prototyping rather than spending hours on a single instance, saving time and effort. Most of the designs included in this portfolio followed such philosophy.

To me, design is the universal language of nature with which we can express our emotion and innovation. It is probably the strongest form of artistic impression an individual can communicate with. Design tells us stories, defines who we are and builds the future we aspire to reach.

Profile.

3+ Years

Experience as a Mechanical Design Engineer

200+

Industrial and Product Designs Completed

130+

Products Delivered as Product Manager

SOLIDWORKS · AUTODESK INVENTOR · AUTODESK FUSION 360 · KeyShot

AUTOCAD · Ansys · SOLIDWORKS Visualize · AUTODESK RECAP

Office · filmora · Figma

CSWA Certification.

DASSAULT SYSTEMES

SOLIDWORKS

CERTIFICATE

Dassault Systèmes confers upon

OMAR SAIF

the certification for

Mechanical Design

at the level of **ASSOCIATE**

July 23 2021

CERTIFIED DASSAULT SYSTEMES ASSOCIATE

Gian Paolo BASSI
CEO SOLIDWORKS

C-9XJZGRMV3T

Scan to Validate

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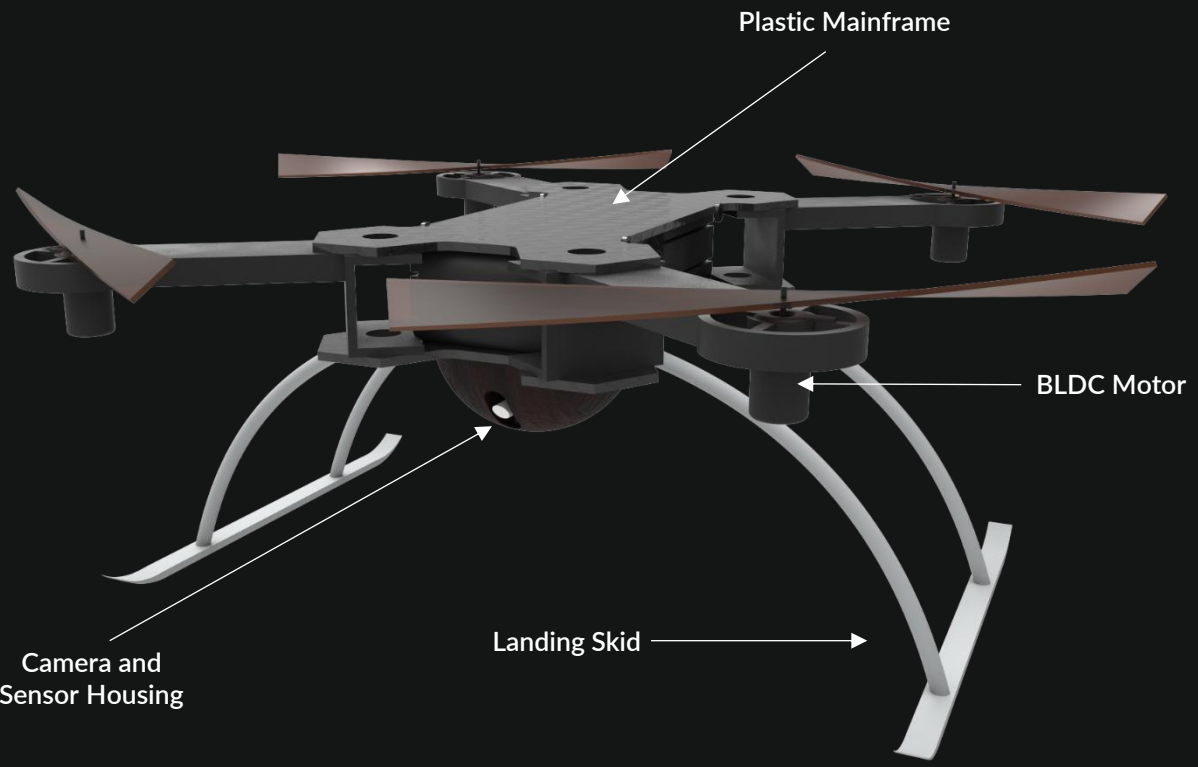
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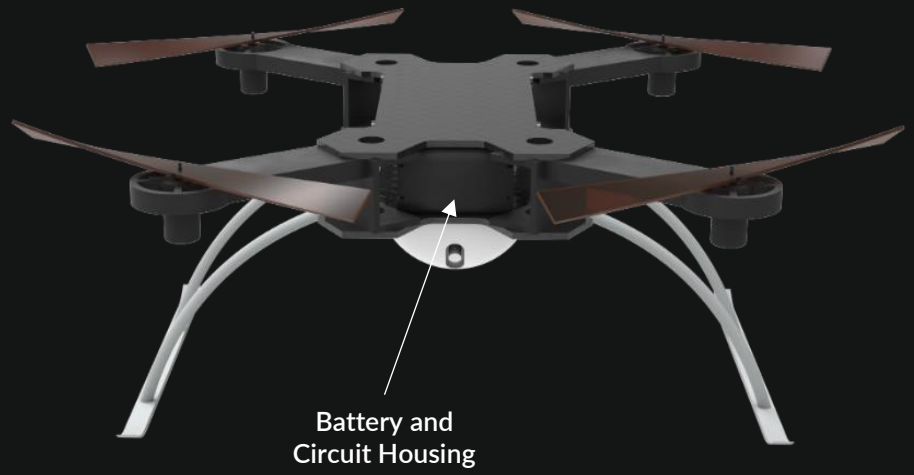
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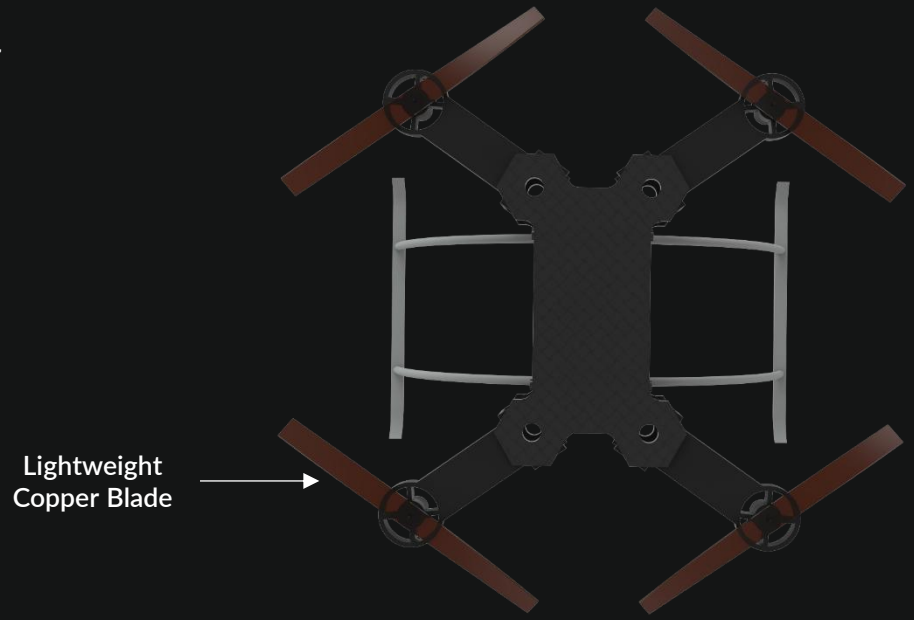
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01 | Mini Quadcopter (concept)

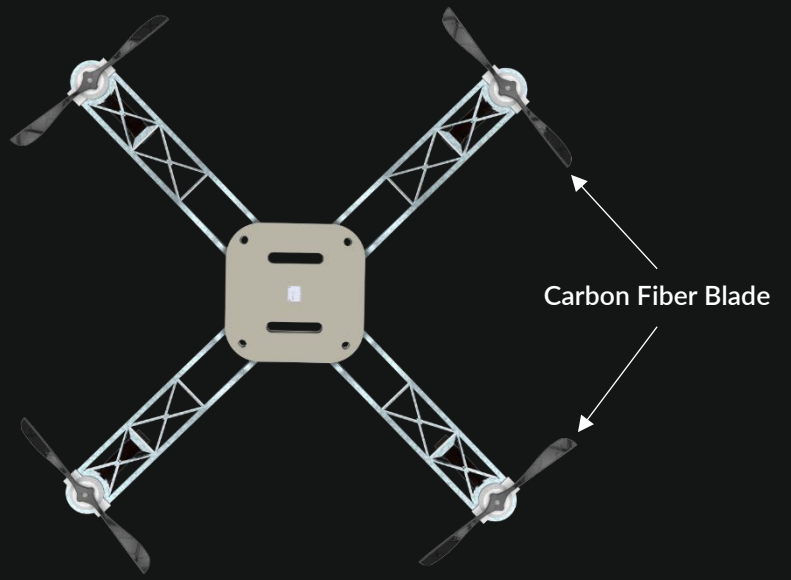


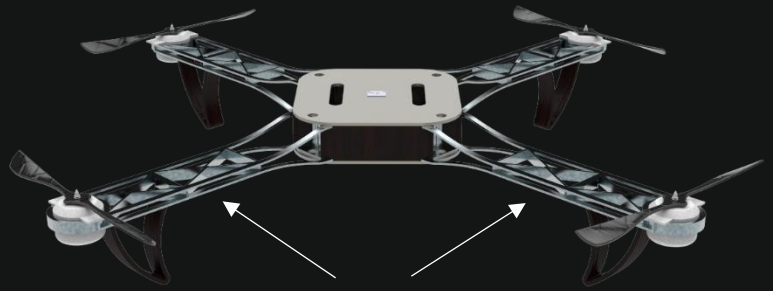




02 | Quadcopter (concept)

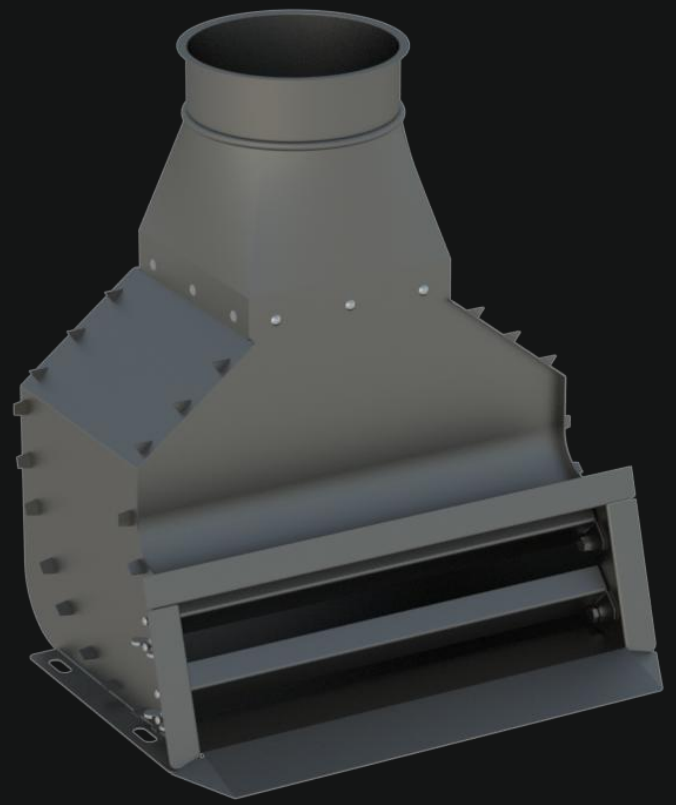


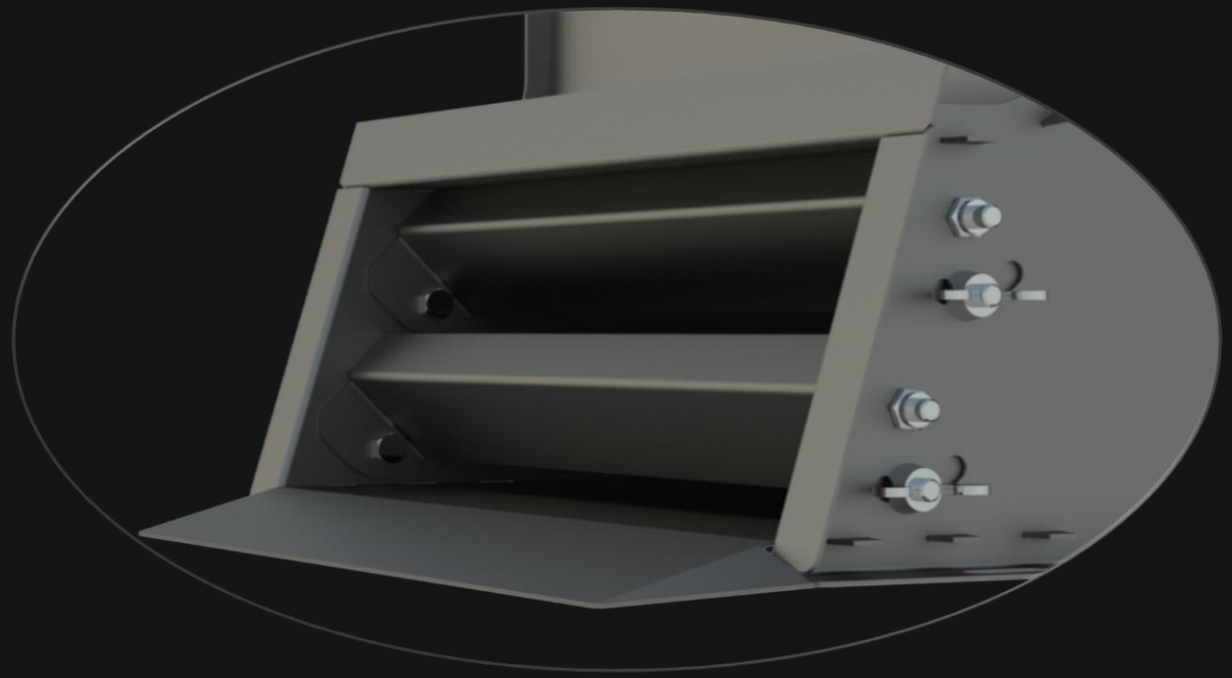






03 | Industrial Vacuum Cleaner



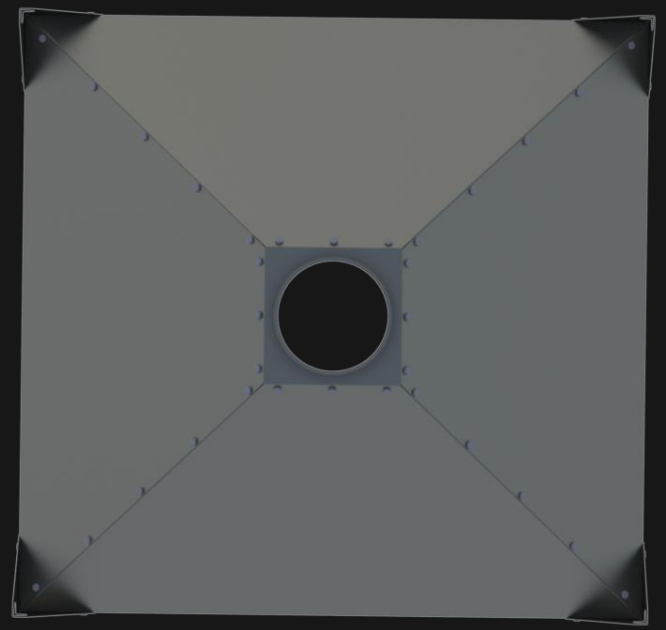


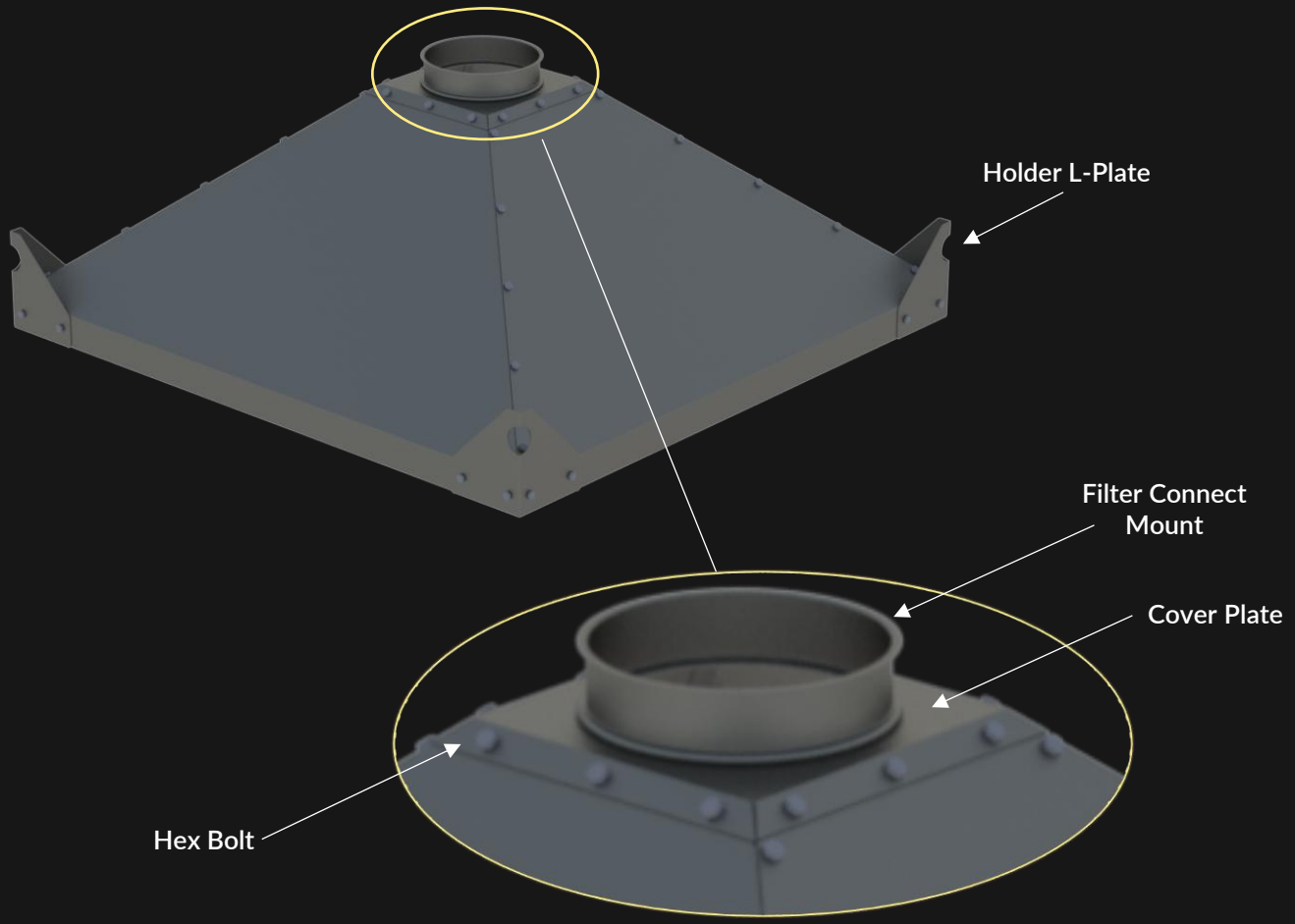
Background

At the time of designing, my company was in a need for a vacuum cleaner for their wood factory. It required a scoop (which is the design shown here) to be connected to the vacuum chamber that can be configured with respect to the flow speed and/or particle size for a given task. As shown in the picture on the right, two flaps were set up with butterfly bolts within an arc like slot. This gave us finer control (as discussed) over the operation of the product.

Background

Our client required a universal exhaust solution that can be used for various materials. This hood was proposed to seal off (and connect to a filter) the chimney associated with that exhaust system for safe and environmentally friendly waste management. With 2.0 mm thick stainless steel, it was designed to tolerate vibration and thermal stresses during the operation.





04 | Chimney Hood

05 | Reading Lamp (concept)



Background

This was a concept developed to play around with some tools in assembly modelling. The model was made in the assembly environment of SolidWorks while Visualize had been used to render the image for presentation purpose.







Key Criteria

- Type : Anti-Material Rifle
- Origin : USA
- Designer : Ronnie Barrette (1980)

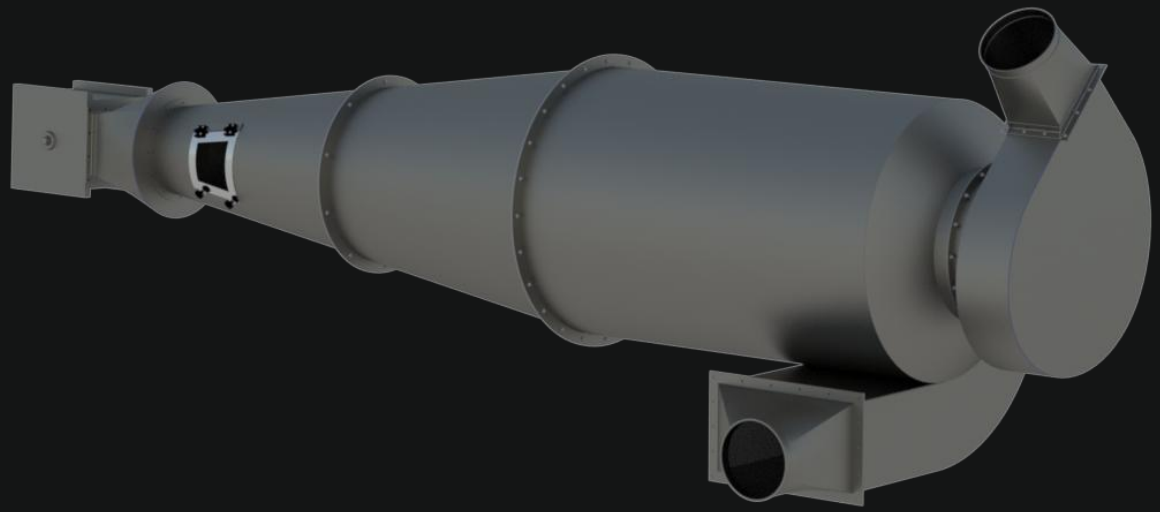
- Mass : 14.8 kg
- Length : 120 – 140 cm
- Muzzle Velocity : 853 m/s

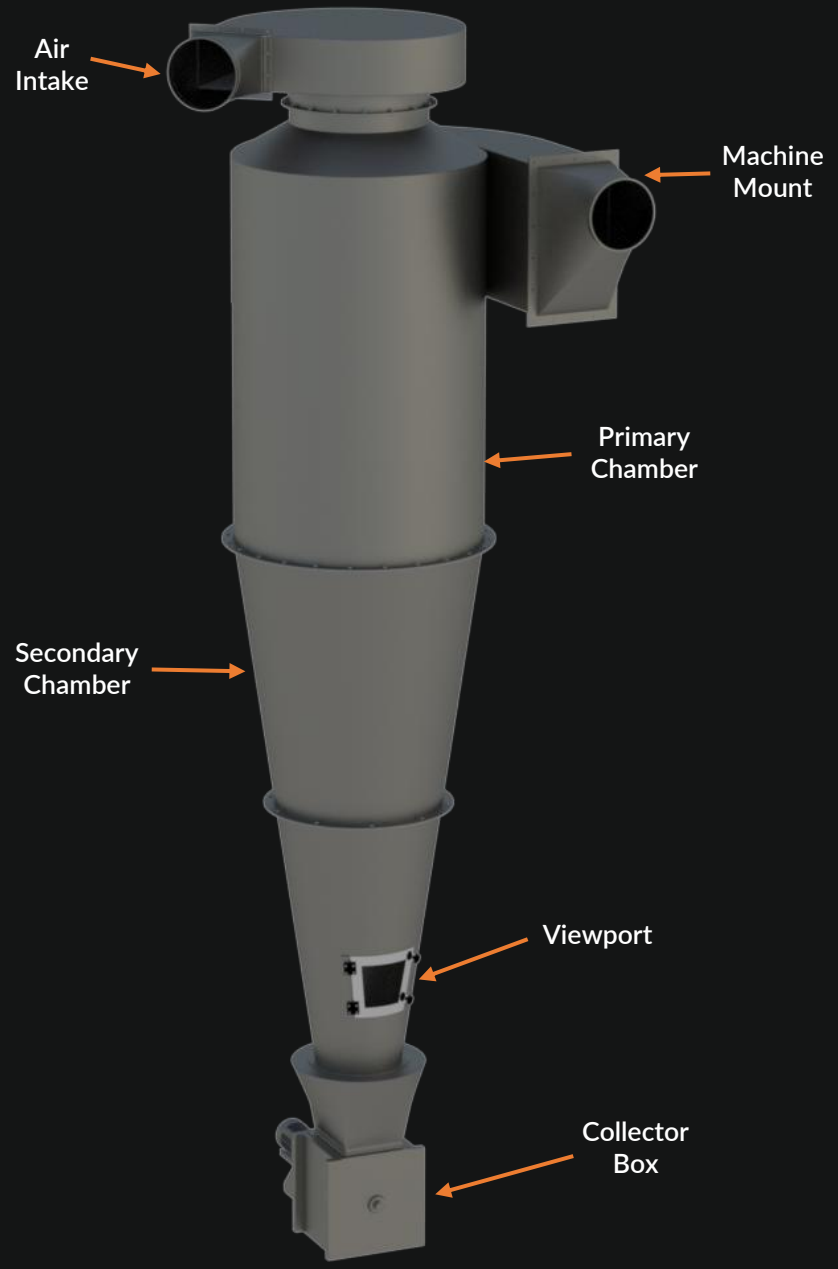
Background

Inspired by popular FPS franchise Call of Duty, this design was a practice project that was entirely done on the SolidWorks environment. No additional tools were used to render. This was also a ‘speed project’ with the aim to finish under 10 hours and was done successfully.

06 | Barrette M82 Sniper Rifle

07 | Dust Extraction Device



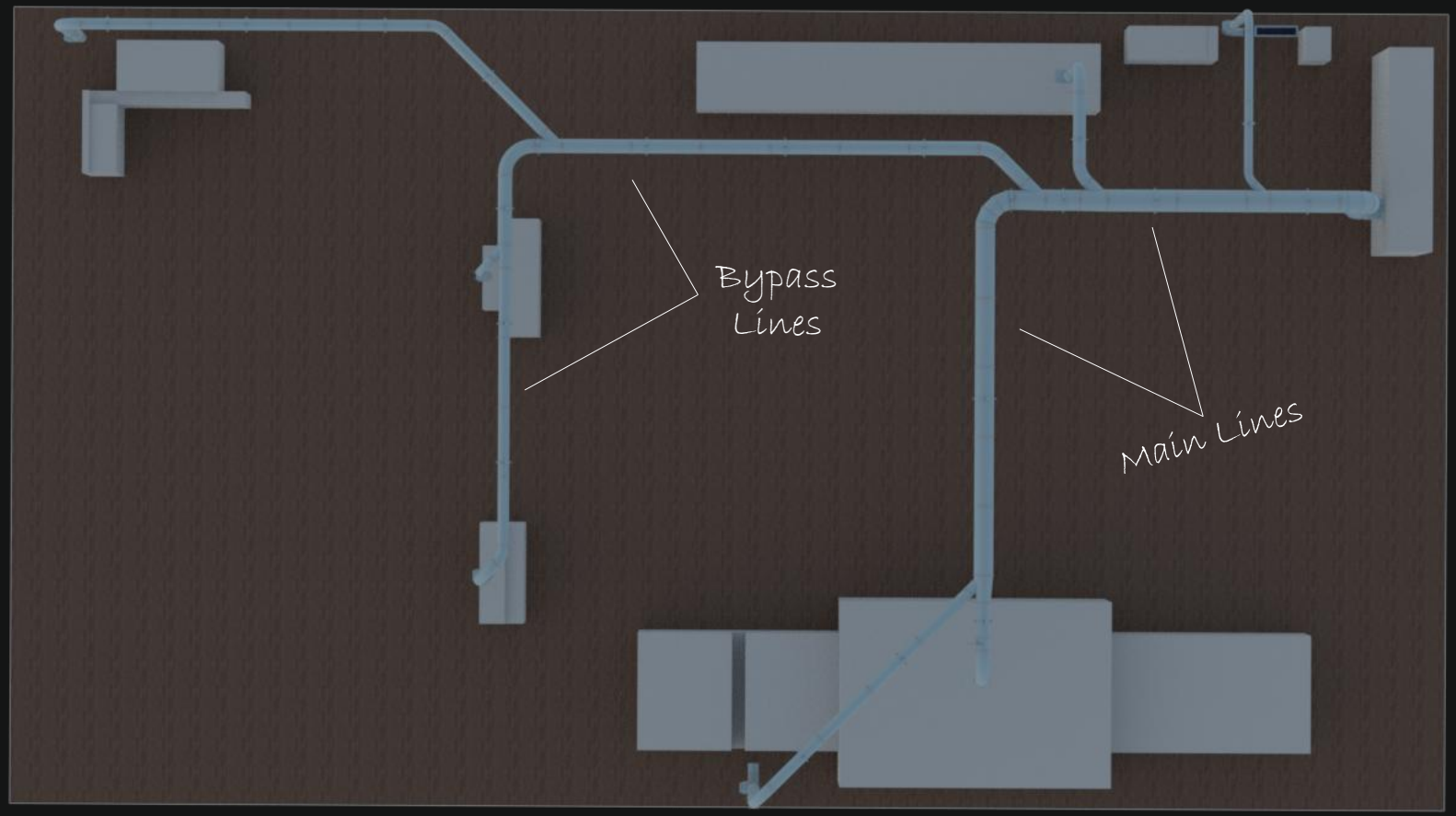


Background

This was designed to be mounted on an wood chipper machine to collect extracts during operation. It was aimed to reduce waste and improve process recyclability.

This particular product was a very difficult task for our team. First of all, the complexity of the design was unlike our other products. Numerous fittings and GD&T had to be taken into consideration. Secondly, it was a large product thus the nesting couldn't be done on a single metal sheet – adding to more challenges. Mountings on a curved surface was also a difficult task to deal with. But at the end, it all turned out as it was planned.

08 | Industrial Piping Layout



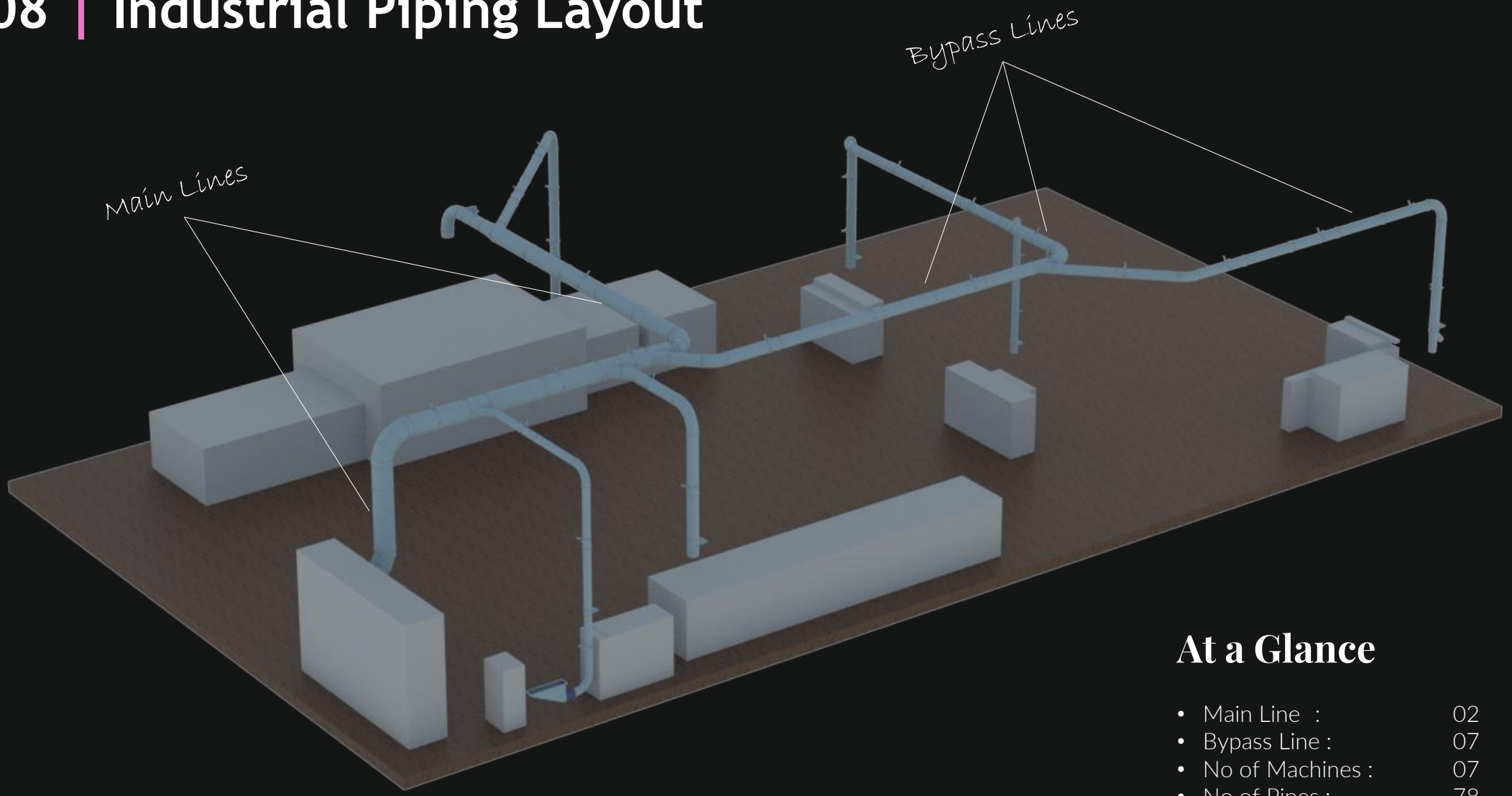
Background

For one of our clients, a comprehensive piping layout (with mountings and accessories) had to be designed for their woodworking plant. Machines had been set up and fluid flow between them had to be taken under consideration besides proper waste management.

This design was aimed to address that from the get go. Using industry standard equipment, I was able to plan, design, and deliver the solution within 5 days.

As various measurements and engineering consideration had to be taken care of from both our side and the client's side, this design was a testament of collaboration and multiple iterations.

08 | Industrial Piping Layout



At a Glance

- Main Line : 02
- Bypass Line : 07
- No of Machines : 07
- No of Pipes : 78

09 | Water Bottle

Background

This was the first experimental design for me on surface modelling. Based on traditional aftermarket sports drink bottle – this concept design was crucial for me to learn the intricate details and versatile tools of surface modelling for organic shape.

Measurements

- Height : 160 mm
- Base : 52 mm (dia)
- Thickness : 1.5 mm (uniform)
- Volume : 1 liter (approx.)



10 | Coffee Maker





Background

3D Model and Rendering of a Coffee Maker with embossed icon of Starbucks. This was an experimental design for me to see various lighting angles and reflection studies. Also, I explored various colors on the same products and see how it affected the final result – both in product colors and background colors.

11 | Café Racer (concept)



